

Composition Pair 10: Knowledge Transfer and DNA Absorption

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This work derives from and formalizes the inter-Self coordination architecture introduced in *Inter-Self Coordination via Shared Substrate / Full Aspect Integration* (Li, April 2026), the third paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026) and *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026).

Abstract

This note formalizes the composition of two Paper 3 operational commitments: Knowledge Transfer FAI Context (D2.64), which governs asymmetric Full Aspect Integration events in which one Self contributes governance patterns for another Self to learn from, and DNA Absorption Governance (D2.11), which governs the five-step protocol through which a Self's home governance directs absorption of DNA-origin content from a shared substrate. When both apply simultaneously, three governance requirements emerge from the combination that neither commitment generates alone. First, the contributor's explicit transfer intent creates implicit absorption pressure that the cross-organizational agreement must explicitly counter by stating that absorption remains the learner's home governance decision. Second, the contributor's contribution records should identify which aspects were contributed with absorption intent and which were contributed for coordination context only, providing the learner with governance intelligence that guides review without creating obligation. Third, the event produces two distinct governance acts — the contributor's transfer record and the learner's absorption records — that must be maintained separately because they carry different accountability principals. Any governed system for AI governance capability transfer must address all three.

1. Pair Identification

Commitment A — Knowledge Transfer FAI Context (D2.64):

A variant of FAI in which the participating Selves hold an asymmetric relationship with respect to governance capability: one Self (the contributor) participates specifically to share governance patterns for the other Self (the learner) to observe, study, and potentially internalize. The contributor's sharing scope is selected with transfer intent — the contributed aspects are chosen because they carry governance patterns the learner is expected to benefit from seeing. The cross-organizational agreement governing the event names this asymmetric purpose. D2.64 is an operational variant of the FAI commitment (Claim 2) and of the cross-organizational agreement commitment (D2.34), applied in the specific governance scenario where capability development is the event's primary purpose rather than task coordination.

Commitment B — DNA Absorption Governance (D2.11):

The five-step governance protocol through which a Self's home governance exercises directed selection over DNA-layer content from a shared substrate. The five steps are: (1) identify candidate DNA content from the shared substrate; (2) assess fitness for the home substrate; (3) record the assessment and governance deliberation; (4) approve or decline; (5) record the final absorption decision with rationale. DNA absorption is an operational variant of the DNA evolution commitment from Paper 2, carried forward through Paper 3's inheritance: the receiving Self's authority structure governs which candidates are reviewed, which are absorbed, and which are declined, and this authority is not diluted by the inter-Self context in which the shared substrate was constructed.

2. Governance Scenario Requiring Both Simultaneously

A more experienced Self, whose home substrate carries mature governance patterns for orchestration, conflict-handling, and schema management, enters an FAI event with a less experienced Self whose home governance has identified governance capability development as a priority. The contributor Self selects a sharing scope chosen specifically because the contributed aspects illustrate the governance patterns the learner needs. The event proceeds under a cross-organizational agreement that names knowledge transfer as the event's purpose.

After the FAI event, the learner Self's governance reviews the shared substrate. The shared substrate now contains the contributor's contributed aspects, including orchestration substrates, governance schemas, and conflict-handling rules that the contributor specifically intended for the learner to consider absorbing. The learner's governance initiates the D2.11 protocol: identifying candidates, assessing fitness, deliberating, deciding, and recording.

Both commitments apply in full. D2.64 governs the event and the contributor's conduct during it. D2.11 governs the learner's absorption decisions after it. The composition is not merely sequential — the knowledge transfer context created in D2.64 actively shapes the governance conditions under which D2.11 must operate, generating requirements that

neither commitment alone anticipates.

3. Non-Obvious Governance Requirements from the Combination

Requirement 1 — Absorption intent must not become absorption obligation.

In a standard bilateral FAI event, no participant has declared a purpose of helping the other absorb their DNA. When the knowledge transfer context is present, the contributor has done exactly that: they have entered the event with an explicit purpose of facilitating absorption. This declared purpose creates implicit governance pressure on the learner. Declining to absorb any candidate from a knowledge transfer event is not a neutral act within the event's social and organizational context — it can read as a rejection of the event's stated purpose, as an implicit judgment that the contributor's governance patterns are inadequate, or as a failure on the learner's part to take the development opportunity seriously.

D2.11 alone does not address this pressure because D2.11 applies to all DNA absorption contexts, most of which carry no such implicit obligation. D2.64 alone does not address this pressure because D2.64 governs the contributor's conduct and the event structure, not the learner's subsequent governance decisions.

The combination requires a specific governance mechanism: the cross-organizational agreement (D2.34) for the event must explicitly state that absorption is the learner's home governance decision and cannot be conditioned on by the contributor. This explicit statement is required specifically because the knowledge transfer context creates implicit absorption expectations that standard FAI events do not. The statement does not merely reaffirm what D2.11 already commits to; it counters a specific social pressure that the composition generates and that would otherwise be invisible to governance review.

Requirement 2 — Absorption-intended scope within sharing scope.

In a knowledge transfer event, the contributor's sharing scope has internal structure that a standard FAI sharing scope does not. Some contributed aspects are intended primarily to give the learner coordination context — enough visibility into how the contributor's substrate operates to make the shared-substrate interaction coherent. Other contributed aspects are specifically chosen because the contributor intends them as absorption candidates: the contributor has identified these as governance patterns the learner's substrate would benefit from absorbing.

The contributor's contribution record (D2.06) should identify which aspects fall into each category. This absorption-intended scope notation serves the learner's D2.11 review: it allows the learner's governance to focus attention on the candidates the contributor considered worth absorbing, rather than treating all contributed aspects as equally strong candidates. Without this notation, the learner's D2.11 review must independently determine which aspects were contributed for absorption value and which were

contributed for context, work that the contributor is better positioned to do and has already done.

The notation must be framed as governance intelligence, not as a mandate. It represents the contributor's intent and assessment, not the learner's commitment. The learner's governance may absorb an aspect not identified as absorption-intended, may decline an aspect explicitly identified as absorption-intended, or may absorb none. The absorption-intended scope notation informs the learner's review; it does not constrain it.

Requirement 3 — Separate governance records for transfer intent and absorption decision.

A knowledge transfer FAI event with DNA absorption produces two distinct governance acts carried out by two distinct accountability principals.

The contributor's knowledge transfer record, governed by D2.64, documents: what aspects were contributed, what transfer purpose each was contributed for, which were identified as absorption-intended, and what governance authorization existed for the contributor's participation. The contributor is accountable for this record. Its accuracy and completeness are the contributor's governance responsibility.

The learner's absorption records, governed by D2.11 Steps 3–5, document: which candidates were reviewed, what the assessment concluded for each, and which were absorbed or declined with rationale. The learner is accountable for these records. Their accuracy and completeness are the learner's governance responsibility.

These two governance acts must not be merged into a single record. A merged record — such as a joint event summary stating that "governance patterns were transferred and absorbed" — conflates the contributor's and learner's accountability, obscures whether the learner's D2.11 protocol was actually executed, and prevents an observer from tracing why specific absorption decisions were made. The learner's absorption records must be independently auditable as the output of the learner's home governance process, not as a joint conclusion of the event.

4. Prior-Art Significance

Any governed system designed to transfer AI governance capability between organizations — whether described as a mentorship protocol, a governance-pattern licensing arrangement, a governance-pattern propagation mechanism, or a capability-development exchange — must address the three requirements this composition identifies.

The first requirement, that transfer intent not become absorption obligation and that this must be stated explicitly in the governing agreement, is a composition-specific property. It is not derivable from the governance of knowledge transfer events alone (which does not address absorption decisions) and not derivable from the governance of DNA absorption

alone (which does not address the social pressure created by explicit transfer intent). The explicit agreement statement is architecturally required by the combination.

The second requirement, that the contributor identify absorption-intended scope separately from context-only contributions, and that this identification be framed as guidance rather than mandate, is also composition-specific. It requires the contributor to perform governance work — classifying the transfer purpose of each contributed aspect — that standard FAI sharing scope notation does not include.

The third requirement, that two separate governance records be maintained for the transfer act and the absorption act respectively, is derivable from the general principle of accountability separation, but its application to this scenario requires recognizing that the knowledge transfer context can create pressure to collapse the two records into one, and that this collapse must be explicitly governed against. Prior art in governance capability transfer must separately address all three to be considered architecturally complete.

5. Operational Test

For a knowledge transfer FAI event in which DNA absorption is also executed, governance is verifiable if and only if an observer can confirm all three of the following:

(a) The cross-organizational agreement for the event explicitly states that absorption is the learner's home governance decision and cannot be conditioned on by the contributor — not merely that governance remains home-governed in general, but that this specific event's knowledge transfer purpose does not create absorption obligation.

(b) The contributor's contribution records identify, for each contributed aspect, whether that aspect was contributed with absorption intent or with coordination-context-only intent, and the identification is framed as the contributor's assessment rather than as a constraint on the learner's review.

(c) The learner's absorption governance records — documenting D2.11 Steps 3 through 5 for each candidate reviewed — exist as independent records attributable to the learner's home governance, separate from the event-level transfer record and auditable without reference to it.

A system that passes all three parts of this test has governed both commitments in their combined form. A system that passes only one or two has governed each commitment individually but has not addressed the governance requirements their composition generates.

Source Papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

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